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# Homework 1 — Indirect Optimization of a Planar Ascent Trajectory

DYNAMICS AND CONTROL OF LAUNCH VEHICLES

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# 1 Introduction

This report presents my solution to Homework 1 of the Dynamics and Control of Launch Vehicles course. The problem is the minimum-fuel planar ascent of a rocket, from rest on the ground to a prescribed low-Earth orbit injection state. I work through four mission programs of increasing complexity, each written as an optimal control problem and solved with the Pontryagin Maximum Principle (PMP) and single shooting over the costate initial conditions. The problem statement and the model follow the course treatment of the planar ascent [1, HW1].

## 1.1 Indirect Optimal Control: Method Overview

All four programs are solved with the indirect approach to optimal control. The idea is to first derive the first-order necessary conditions of optimality analytically, from the Pontryagin Maximum Principle [2], and only solve the resulting boundary value problem numerically afterwards. Figure 1 shows the whole pipeline, from the analytical PMP steps that reduce the problem to a boundary value problem (BVP) down to the MATLAB routines that actually solve it; the same scheme is used for all four tasks.

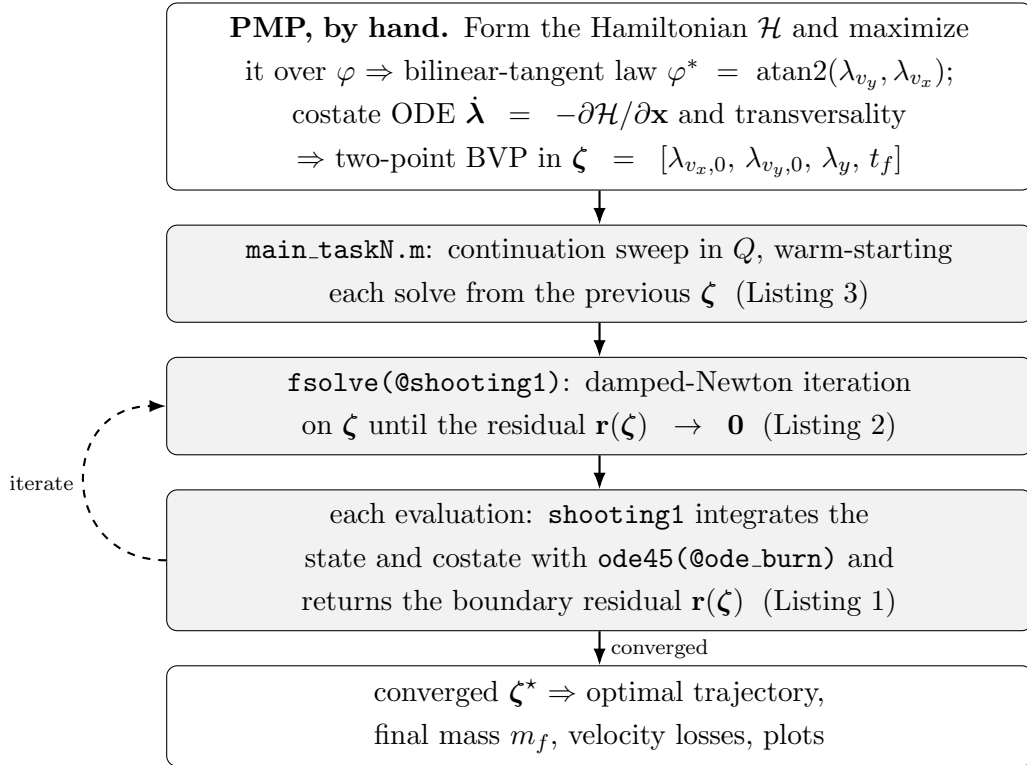


Figure 1: Solution pipeline for all four tasks: the analytical PMP steps (top) reduce the problem to a BVP, which the MATLAB routines (shaded) solve by continuation and single shooting. Listing numbers refer to Appendix B.

The alternative is the direct approach, which goes the other way: it discretizes

the trajectory first and then optimizes, turning the problem into a large nonlinear program [3, Sec. 4.12.7]. I went with the indirect route mainly because it is accurate, since the optimality conditions are imposed exactly rather than approximated on a mesh, and because it keeps a clear physical meaning: the velocity costate  $\lambda_v$  is the primer vector [1, Lecture 6] that sets the optimal thrust direction. It also cuts the whole problem down to just a few unknowns. The catch is that the shooting residual is very sensitive to the costate guess, which has no physical meaning; the convergence basins are small, and you have to guess the thrust-arc (switching) structure in advance. Section 1.5 comes back to these issues and to the continuation strategy I used to handle them.

## 1.2 Vehicle Dynamics

The vehicle is modelled as a point mass moving in a vertical plane above a flat, non-rotating Earth with no atmospheric drag. The state vector is

$$\mathbf{x}(t) = [x, y, v_x, v_y, m]^T, \quad (1)$$

where  $(x, y)$  is the position in the orbital plane,  $(v_x, v_y)$  is the velocity, and  $m$  is the vehicle mass. The single control variable is the thrust-direction angle  $\varphi$  measured from the horizontal. The thrust magnitude is  $T = cQ$ , where  $c$  is the effective exhaust velocity and  $Q = -\dot{m}$  is the constant mass-flow rate. The equations of motion are

$$\dot{x} = v_x, \quad (2)$$

$$\dot{y} = v_y, \quad (3)$$

$$\dot{v}_x = \frac{T}{m} \cos \varphi, \quad (4)$$

$$\dot{v}_y = \frac{T}{m} \sin \varphi - g, \quad (5)$$

$$\dot{m} = -Q. \quad (6)$$

Figure 2 shows the planar geometry of the problem, as given in the assignment.

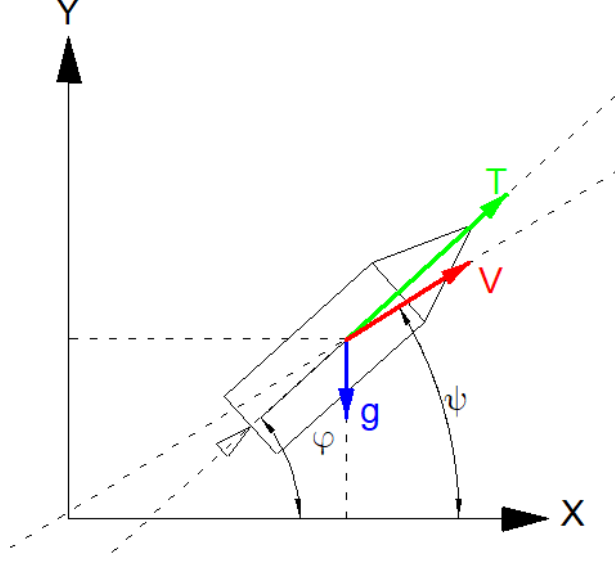


Figure 2: Planar ascent geometry from the assignment statement. The thrust  $\mathbf{T}$  (magnitude  $T = cQ$ ) acts along the vehicle axis at pitch angle  $\varphi$  from the horizontal; the velocity  $\mathbf{V}$  makes the flight-path angle  $\psi$  with the horizontal; gravity  $\mathbf{g}$  points down. The thrust components in Eqs. (4)–(5) are  $T \cos \varphi$  and  $T \sin \varphi$ .

### 1.3 Non-dimensionalization

All quantities are made dimensionless with the reference values in Table 1. The reference velocity  $V_{\text{ref}}$  is the circular orbital speed at the nominal target altitude; the reference length and time follow from  $g$ .

Table 1: Non-dimensionalization reference values.

| Quantity     | Symbol           | Expression           | Value                 |
|--------------|------------------|----------------------|-----------------------|
| Velocity     | $V_{\text{ref}}$ | orbital speed        | 7800 m/s              |
| Acceleration | $g_{\text{ref}}$ | $g$                  | 9.81 m/s <sup>2</sup> |
| Length       | $L_{\text{ref}}$ | $V_{\text{ref}}^2/g$ | $\approx 6200$ km     |
| Time         | $t_{\text{ref}}$ | $V_{\text{ref}}/g$   | $\approx 795$ s       |
| Mass         | $m_{\text{ref}}$ | $m_0$                | $10^6$ kg             |

In non-dimensional form ( $g = 1$ ,  $m_0 = 1$ ,  $v_{x,f} = 1$ ) the fixed problem data are collected in Table 2.

Table 2: Fixed non-dimensional problem parameters.

| Symbol | Value            | Physical interpretation                                                                    |
|--------|------------------|--------------------------------------------------------------------------------------------|
| $c$    | 0.6              | Effective exhaust velocity ( $c_{\text{dim}} \approx 4680$ m/s,<br>$I_{sp} \approx 477$ s) |
| $\eta$ | 0.1              | Structural coefficient $m_s/m_p$                                                           |
| $y_f$  | 0.04, 0.05, 0.06 | Target orbit altitude ( $\approx 248, 310, 372$ km)                                        |

The mass-flow rate  $Q$  is a free design parameter swept in Tasks 1 and 4. Since  $W = mg$  and  $m_0 = 1$ , the initial thrust-to-weight ratio is  $T/W|_{t=0} = cQ$ , so liftoff ( $T > W$ ) requires  $Q > 1/c \approx 1.67$ . As propellant burns off ( $\dot{m} = -Q < 0$ ) the ratio  $T/W = cQ/m$  increases monotonically, hence the liftoff instant is the binding case.

The value  $Q = 2$  is the nominal mass-flow rate given in the assignment. It corresponds to a liftoff thrust-to-weight ratio  $T/W|_{t=0} = cQ = 1.2$ , which is representative of a real first stage and safely above the  $Q_{\min} = 1/c \approx 1.67$  liftoff threshold. Task 1 sweeps  $Q$  to study its effect and finds the payload-optimal rate  $Q^* \approx 2.2$ – $2.5$  (Table 3), so the nominal value sits just below the single-stage optimum. Tasks 2–4 then keep  $Q = 2$  fixed, so that the comparison of the structural features (vertical climb, coast, staging) is made at a single, common operating point.

## 1.4 Pontryagin Maximum Principle

### 1.4.1 Hamiltonian and Costate Equations

The optimal control problem is to find the control  $\varphi(t)$  and the free terminal time  $t_f$  that maximize the final mass  $m(t_f)$ . In Mayer form, the terminal (Mayer) cost is  $\phi = m(t_f)$  and, since there is no running term, the cost functional is simply  $J = \phi$ ; we maximize  $J = \phi$  subject to the dynamics (2)–(6) and the terminal constraints

$$y(t_f) = y_f, \quad v_x(t_f) = 1, \quad v_y(t_f) = 0. \quad (7)$$

The horizontal position  $x(t_f)$  is free, and  $t_f$  is free.

Introducing the costate (adjoint) vector  $\boldsymbol{\lambda} = [\lambda_x, \lambda_y, \lambda_{v_x}, \lambda_{v_y}, \lambda_m]^T$ , the Hamiltonian is

$$\mathcal{H} = \lambda_x v_x + \lambda_y v_y + \lambda_{v_x} \frac{T}{m} \cos \varphi + \lambda_{v_y} \left( \frac{T}{m} \sin \varphi - 1 \right) - \lambda_m Q. \quad (8)$$

Rearranging,

$$\mathcal{H} = \underbrace{\lambda_x v_x + \lambda_y v_y - \lambda_{v_y} - \lambda_m Q}_{\text{state-only part}} + \frac{T}{m} \underbrace{(\lambda_{v_x} \cos \varphi + \lambda_{v_y} \sin \varphi)}_{\text{control-dependent part}}. \quad (9)$$

The Euler–Lagrange costate equations  $\dot{\boldsymbol{\lambda}} = -\partial\mathcal{H}/\partial\mathbf{x}$  [4, Ch. 2] yield

$$\dot{\lambda}_x = 0 \quad \Longrightarrow \quad \lambda_x = 0 \quad (x(t_f) \text{ free} \Rightarrow \lambda_x(t_f) = 0), \quad (10)$$

$$\dot{\lambda}_y = 0 \quad \Longrightarrow \quad \lambda_y = \text{const}, \quad (11)$$

$$\dot{\lambda}_{v_x} = -\lambda_x = 0 \quad \Longrightarrow \quad \lambda_{v_x} = \lambda_{v_x,0} = \text{const}, \quad (12)$$

$$\dot{\lambda}_{v_y} = -\lambda_y \quad \Longrightarrow \quad \lambda_{v_y}(t) = \lambda_{v_y,0} - \lambda_y t, \quad (13)$$

$$\dot{\lambda}_m = \frac{T}{m^2} \|\boldsymbol{\lambda}_v\|, \quad (14)$$

where  $\|\boldsymbol{\lambda}_v\| = \sqrt{\lambda_{v_x}^2 + \lambda_{v_y}^2}$ .

Four of the five costate components are determined analytically:  $\lambda_x = 0$  identically,  $\lambda_y$  and  $\lambda_{v_x}$  are constant, and  $\lambda_{v_y}$  varies linearly in time. Only  $\lambda_m$  requires numerical integration via Eq. (14). This analytical structure cuts the cost of evaluating the shooting residual.

#### 1.4.2 Optimal Control Law

By the PMP, the optimal control  $\varphi^*$  maximizes  $\mathcal{H}$  pointwise. Since  $T/m > 0$ , this is equivalent to maximizing the control-dependent part of Eq. (8),

$$g(\varphi) = \lambda_{v_x} \cos \varphi + \lambda_{v_y} \sin \varphi, \quad (15)$$

over  $\varphi$ . The first-order stationarity condition reads

$$\frac{\partial \mathcal{H}}{\partial \varphi} = \frac{T}{m} (-\lambda_{v_x} \sin \varphi + \lambda_{v_y} \cos \varphi) = 0 \quad \Longrightarrow \quad \tan \varphi^* = \frac{\lambda_{v_y}(t)}{\lambda_{v_x}}. \quad (16)$$

This is the **bilinear-tangent (linear-tangent) law** [4, Sec. 2.4]: since  $\lambda_{v_y}$  is linear in  $t$  (Eq. 13) and  $\lambda_{v_x}$  is constant,  $\tan \varphi^*$  varies linearly with time.

**Geometric interpretation and branch selection.** Equation (16) determines  $\varphi^*$  only up to an additive  $\pi$ : it admits two roots per period. Geometrically,  $\lambda_{v_x}$  and  $\lambda_{v_y}$  are the two legs of a right triangle whose hypotenuse is the norm  $\|\boldsymbol{\lambda}_v\| = \sqrt{\lambda_{v_x}^2 + \lambda_{v_y}^2}$ , so the two candidate directions are

$$(I) \quad \cos \varphi^* = \frac{\lambda_{v_x}}{\|\boldsymbol{\lambda}_v\|}, \quad \sin \varphi^* = \frac{\lambda_{v_y}}{\|\boldsymbol{\lambda}_v\|} \quad \text{or} \quad (II) \quad \cos \varphi^* = -\frac{\lambda_{v_x}}{\|\boldsymbol{\lambda}_v\|}, \quad \sin \varphi^* = -\frac{\lambda_{v_y}}{\|\boldsymbol{\lambda}_v\|}. \quad (17)$$

Substituting each branch into Eq. (15) discriminates between them:

$$g_I = \frac{\lambda_{v_x}^2 + \lambda_{v_y}^2}{\|\boldsymbol{\lambda}_v\|} = \frac{\|\boldsymbol{\lambda}_v\|^2}{\|\boldsymbol{\lambda}_v\|} = +\|\boldsymbol{\lambda}_v\|, \quad g_{II} = -\|\boldsymbol{\lambda}_v\|. \quad (18)$$

Branch (I) yields the maximum and branch (II) the minimum, so the PMP selects branch (I): the optimal thrust direction is aligned with the velocity costate  $\boldsymbol{\lambda}_v$  (primer vector), and the control-dependent term attains its largest value,

$$\lambda_{v_x} \cos \varphi^* + \lambda_{v_y} \sin \varphi^* = \|\boldsymbol{\lambda}_v\|. \quad (19)$$

Using this result, the optimal Hamiltonian simplifies to

$$\mathcal{H}^* = \lambda_y v_y + \frac{T}{m} \|\boldsymbol{\lambda}_v\| - \lambda_{v_y} - \lambda_m Q. \quad (20)$$

### 1.4.3 Transversality Conditions

The costate boundary conditions follow from the transversality (natural boundary) conditions of the PMP [4, Ch. 2]. Let  $\phi = m(t_f)$  denote the terminal (Mayer) cost, which here is also the whole objective  $J$  since there is no running term, and adjoin the terminal constraints  $\boldsymbol{\chi}(\mathbf{x}(t_f)) = [y - y_f, v_x - 1, v_y]^T = \mathbf{0}$  with multipliers  $\boldsymbol{\mu}$ . The augmented cost functional is

$$\bar{J} = \phi + \boldsymbol{\mu}^T \boldsymbol{\chi}(\mathbf{x}(t_f)) + \int_{t_0}^{t_f} \boldsymbol{\lambda}^T (\mathbf{f} - \dot{\mathbf{x}}) dt, \quad (21)$$

whose integral vanishes on any feasible trajectory ( $\dot{\mathbf{x}} = \mathbf{f}$ ) and therefore leaves the optimum unchanged. Integrating the  $\boldsymbol{\lambda}^T \dot{\mathbf{x}}$  contribution by parts and collecting the terms evaluated at  $t_f$ , the first variation of  $\bar{J}$  accumulates the terminal boundary term

$$\delta \bar{J}|_{t_f} = \left[ \frac{\partial \phi}{\partial \mathbf{x}_f} + \left( \frac{\partial \boldsymbol{\chi}}{\partial \mathbf{x}_f} \right)^T \boldsymbol{\mu} - \boldsymbol{\lambda}(t_f) \right]^T \delta \mathbf{x}_f, \quad (22)$$

which must vanish for arbitrary admissible  $\delta \mathbf{x}_f$ . This yields the transversality condition

$$\boldsymbol{\lambda}(t_f) = \frac{\partial \phi}{\partial \mathbf{x}_f} + \left( \frac{\partial \boldsymbol{\chi}}{\partial \mathbf{x}_f} \right)^T \boldsymbol{\mu}. \quad (23)$$

Each terminal state falls into one of three cases [1, Lecture 5]. The constrained states  $y, v_x, v_y$  leave their costates  $\lambda_y(t_f), \lambda_{v_x}(t_f), \lambda_{v_y}(t_f)$  undetermined: these absorb the unknown multipliers  $\boldsymbol{\mu}$  and are carried as shooting unknowns (Eq. 27). The horizontal position  $x$  is free and does not appear in the objective, so  $\partial \phi / \partial x = 0$  and Eq. (23) returns  $\lambda_x(t_f) = 0$ , consistent with  $\lambda_x \equiv 0$  already obtained in Eq. (10). The terminal mass  $m(t_f)$  is free but does appear in the objective, with  $\partial \phi / \partial m = 1$ , and this fixes the only nontrivial terminal costate:

$$\lambda_m(t_f) = \frac{\partial \phi}{\partial m(t_f)} = 1. \quad (24)$$

This condition does two things. First,  $\lambda_m(t_f)$  is the sensitivity of the optimal objective to the terminal mass: a unit increase in  $m(t_f)$  improves it by one unit. Second, the PMP only determines  $\boldsymbol{\lambda}$  up to a positive scale factor (the Hamiltonian is homogeneous of degree one in  $\boldsymbol{\lambda}$ ), so Eq. (24) also fixes the normalization of the costate vector.<sup>1</sup>

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<sup>1</sup>Here  $\phi$  is the terminal (Mayer) cost and  $\boldsymbol{\chi}, \boldsymbol{\mu}$  the boundary-constraint vector and its multipliers. In the general Bolza cost  $J = \phi + \int \Phi dt$ , the Mayer term is  $\phi$  and the running (Lagrange) term



Because the terminal time  $t_f$  is free, the PMP supplies one further condition—the natural boundary condition for  $t_f$  [4, Ch. 2]:

$$\mathcal{H}(t_f) = 0. \quad (25)$$

The system is autonomous (no explicit time dependence), so the Hamiltonian is constant along an optimal trajectory;  $\mathcal{H}(t_f) = 0$  therefore implies  $\mathcal{H}(t) \equiv 0$  throughout the powered arc, consistent with the optimality conditions.

## 1.5 Shooting Method and Convergence

Substituting the bilinear-tangent law (16) into Eqs. (2)–(6), the optimal control problem reduces to a **Boundary Value Problem (BVP)** solved by single shooting over the unknown costate initial conditions and the free final time.

I used **two small numerical tricks** to keep the BVP compact and well conditioned.

**(i) Costate normalization.** Since the Hamiltonian is homogeneous of degree one in  $\boldsymbol{\lambda}$ , the costates are defined only up to a positive scale; we fix it by setting  $\lambda_{m,0} = 1$ , which removes  $\lambda_{m,0}$  as an unknown together with the terminal condition  $\lambda_m(t_f) = 1$ .

**(ii) Hamiltonian imposed at  $t_0$ .** Since the problem is autonomous,  $\mathcal{H}$  is constant and equal to zero (free  $t_f$ ); we therefore impose  $\mathcal{H} = 0$  at the initial instant, rather than at  $t_f$  where it would accumulate integration error. Evaluating the optimal Hamiltonian (20) at  $t_0$  is purely algebraic: the launch state has  $v_x = v_y = 0$ , so the term  $\lambda_y v_y$  drops; the initial mass is  $m_0 = 1$ , so  $T/m = T$ ; and with the normalization  $\lambda_{m,0} = 1$  together with  $Q = T/c$  at full thrust, the propellant term becomes  $-\lambda_m Q = -T/c$ . Collecting these contributions,

$$\mathcal{H}(0) = -\lambda_{v_y,0} + T \|\boldsymbol{\lambda}_{v,0}\| - \frac{T}{c} = -\lambda_{v_y,0} + T \left( \|\boldsymbol{\lambda}_{v,0}\| - \frac{1}{c} \right) = 0, \quad (26)$$

where the bracket is the initial value of the switching function  $S = \|\boldsymbol{\lambda}_v\|/m - \lambda_m/c$  (Section 4), evaluated at  $m_0 = 1$  and  $\lambda_{m,0} = 1$ . With  $\lambda_x \equiv 0$  (Eq. 10) already enforced, the shooting unknowns reduce to four,

$$\boldsymbol{\zeta} = [\lambda_{v_x,0}, \lambda_{v_y,0}, \lambda_y, t_f]^T, \quad (27)$$

and  $\lambda_m$  no longer enters the residual. Given a trial  $\boldsymbol{\zeta}$ , the state  $[x, y, v_x, v_y, m]^T$  is integrated forward with `ode45` (RelTol =  $10^{-10}$ , AbsTol =  $10^{-12}$ ), using the right-hand side of Listing 1, and the four residuals are driven to zero with `fsolve` (tolerances

---

$\Phi$  enters the Hamiltonian  $\mathcal{H} = \Phi + \boldsymbol{\lambda}^\top \mathbf{f}$ ; here  $\Phi \equiv 0$  (pure Mayer) [1, Lecture 7]. The sign of the terminal costate depends on the convention used to assemble the augmented functional: the  $(\mathbf{f} - \dot{\mathbf{x}})$  form adopted in Eq. (21), consistent with the maximization of  $\mathcal{H}$ , gives  $\lambda_m(t_f) = +1$  and  $\dot{\lambda}_m = (T/m^2)\|\boldsymbol{\lambda}_v\| \geq 0$  along a powered arc (with equality only where  $T = 0$ ); the opposite  $(\dot{\mathbf{x}} - \mathbf{f})$  form would carry a global minus sign. Both conventions yield the same physical solution; the  $+1$  normalization is the one used in the shooting residual and in `ode.burn.m`.

$10^{-10}$ ). This residual is coded in the function `shooting1` (Listing 2). The same four-unknown formulation is used for the powered arcs of Tasks 1, 2 and 4; Task 3 keeps  $\lambda_m$  as a fifth unknown because the engine-cutoff (switching) condition of the coast needs it.

**Shooting function and convergence.** The residuals define the shooting function  $\mathbf{S} : \boldsymbol{\zeta} \mapsto \mathbf{r}(\boldsymbol{\zeta})$ , the mismatch of the forward-integrated trajectory against the boundary conditions, which `fsolve` drives to zero by a damped Newton iteration

$$\boldsymbol{\zeta}^{k+1} = \boldsymbol{\zeta}^k - \gamma_k \left[ \frac{\partial \mathbf{S}}{\partial \boldsymbol{\zeta}} \right]^{-1} \mathbf{S}(\boldsymbol{\zeta}^k), \quad (28)$$

with a finite-difference Jacobian (propagating the state-transition matrix would be the analytic alternative) [1, Lecture 5]. Single shooting is hard here because the unknowns are costates with no physical meaning: a small change in  $\boldsymbol{\lambda}_0$  gets amplified by the forward integration into a large change in the terminal residual, so the Jacobian is ill-conditioned, the convergence basin is small, and a cold start usually diverges. The problem gets worse when a switching structure is present (Task 3), where the shooting function can even become non-differentiable [1, Lecture 6].

The **continuation (homotopy) strategy** is the standard fix (Listing 3): the BVP is first solved at a favorable operating point ( $Q \approx 3$ ), and each subsequent value of  $Q$  is started from the converged solution at the neighboring point, sweeping both forward and backward across the range  $Q \in [1.8, 7]$ . By embedding the family of problems in the single parameter  $Q$ , every solve gets an initial guess that is already inside the local basin, so the solution branch is followed smoothly without manual retuning. The arc-splitting used in Tasks 2–4 (vertical climb, coast, staging) does a similar job: it breaks a stiff single shoot into better-conditioned sub-arcs, which is the indirect-method version of multiple shooting [1, Lecture 5].

## 2 Task 1: Single Burn Arc

### 2.1 Problem Statement

The simplest mission profile consists of a single continuous powered arc starting from rest at the origin,  $(x, y, v_x, v_y) = (0, 0, 0, 0)$  with  $m(0) = 1$ , and ending at the orbital injection state satisfying Eq. (7). The mass-flow rate  $Q$  is the primary free parameter swept over the range  $Q \in [1.8, 7]$ ; values below  $Q_{\min} = 1/c \approx 1.67$  would yield insufficient thrust at liftoff ( $T/W|_{t=0} = cQ < 1$ ).

### 2.2 Boundary Value Problem Formulation

The four shooting unknowns  $\zeta$  (Eq. 27) determine the costate initial conditions and the free flight time. The shooting residual, coded in the function `shooting1` of `main_task1.m` (Listing 2), is

$$\mathbf{r}_1(\zeta) = \begin{bmatrix} y(t_f) - y_f \\ v_x(t_f) - 1 \\ v_y(t_f) \\ \mathcal{H}(0) \end{bmatrix} = \mathbf{0}. \quad (29)$$

The first three components enforce the terminal state constraints (7); the fourth is the free-time condition  $\mathcal{H} = 0$ , imposed at  $t_0$  through the algebraic form (26). The mass-costate transversality  $\lambda_m(t_f) = 1$  and the terminal value of  $\mathcal{H}$  are no longer needed as residuals, having been folded into the  $\lambda_{m,0} = 1$  normalization and the autonomy of the system.

### 2.3 Velocity-Loss Decomposition

The achieved injection speed falls short of the ideal Tsiolkovsky delta-V,

$$\Delta V_{\text{ideal}} = c \ln \frac{m_0}{m_f} = c \ln \frac{1}{m_f} = \int_0^{t_f} \frac{T}{m} dt, \quad (30)$$

where the last equality follows from the mass dynamics  $\dot{m} = -T/c$ . The shortfall is the total velocity loss  $W_{\text{tot}} = \Delta V_{\text{ideal}} - v_f$ , with  $v_f = 1$ . To resolve it into physical contributions, project the equations of motion onto the velocity unit vector: with flight-path angle  $\psi = \arctan(v_y/v_x)$  and thrust angle  $\varphi$ , the speed evolves as

$$\frac{d\|\mathbf{v}\|}{dt} = \frac{T}{m} \cos(\varphi - \psi) - \sin \psi, \quad (31)$$

the first term being the thrust projected along the velocity and the second gravity (non-dimensional  $g = 1$ ) projected along it. Integrating from rest ( $\|\mathbf{v}\|_0 = 0$ ) to

injection ( $\|\mathbf{v}\|_{t_f} = v_f$ ) and subtracting the result from  $\Delta V_{\text{ideal}}$  in Eq. (30) yields the exact identity

$$\underbrace{\Delta V_{\text{ideal}} - v_f}_{W_{\text{tot}}} = \underbrace{\int_0^{t_f} \frac{T}{m} [1 - \cos(\varphi - \psi)] dt}_{W_d} + \underbrace{\int_0^{t_f} \sin \psi dt}_{W_g}, \quad (32)$$

which splits the loss into a thrust-misalignment term and a gravity term,

$$W_d = \int_0^{t_f} \frac{T}{m} [1 - \cos(\varphi - \psi)] dt \quad (\text{thrust misalignment loss}), \quad (33)$$

$$W_g = \int_0^{t_f} \sin \psi dt \quad (\text{gravity loss}). \quad (34)$$

The misalignment loss  $W_d$  collects the fraction of thrust not aligned with the velocity vector (it vanishes only when  $\varphi \equiv \psi$ ); the gravity loss  $W_g$  is the component of gravity opposing the motion, integrated along the trajectory.

## 2.4 Results

Figure 3 shows the final mass as a function of  $Q$  for the three target altitudes, obtained by solving the BVP at each  $Q$  with the continuation sweep of `main.task1.m` (Listing 3). Each curve has a clear interior maximum  $Q^*$ : at low  $Q$  the long burn time builds up large gravity losses, while at high  $Q$  the vehicle climbs steeply and the thrust-misalignment loss takes over.

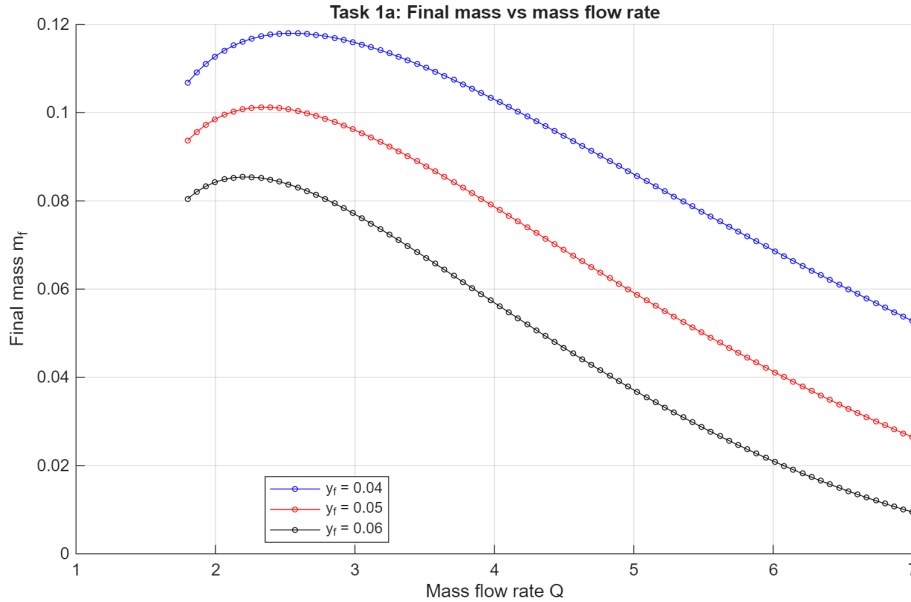


Figure 3: Task 1a: final mass  $m_f$  as a function of the mass-flow rate  $Q$  for three target altitudes  $y_f$ .

The interior maxima are listed in Table 3: the optimal rate  $Q^*$  decreases from 2.52 to 2.19 as the target altitude rises from  $y_f = 0.04$  to 0.06, and the delivered payload

becomes negative at  $y_f = 0.06$ . A single stage cannot reach that orbit with positive payload, which is what motivates the staging in Task 4. The nominal  $Q = 2$  of the assignment sits a bit below  $Q^*$ , but the  $m_f(Q)$  curve is flat near its peak, so the final mass at  $Q = 2$  is still within a few percent of the optimum.

Table 3: Payload-optimal mass-flow rate  $Q^*$  (maximizing the final mass) for the three target altitudes, read from the  $Q$ -sweep of Figure 3. The delivered single-stage payload is  $m_u = m_f(1 + \eta) - \eta$ .

| $y_f$ | $Q^*$ | $m_f^*$ | $m_u^*$ |
|-------|-------|---------|---------|
| 0.04  | 2.52  | 0.1180  | +0.0298 |
| 0.05  | 2.33  | 0.1012  | +0.0114 |
| 0.06  | 2.19  | 0.0854  | −0.0060 |

Figure 4 confirms this trade-off by decomposing the velocity loss at  $y_f = 0.04$ . The gravity loss  $W_g$  is large at low  $Q$  and decreases monotonically with  $Q$ , while the misalignment loss  $W_d$  has a broad minimum near  $Q^*$ .

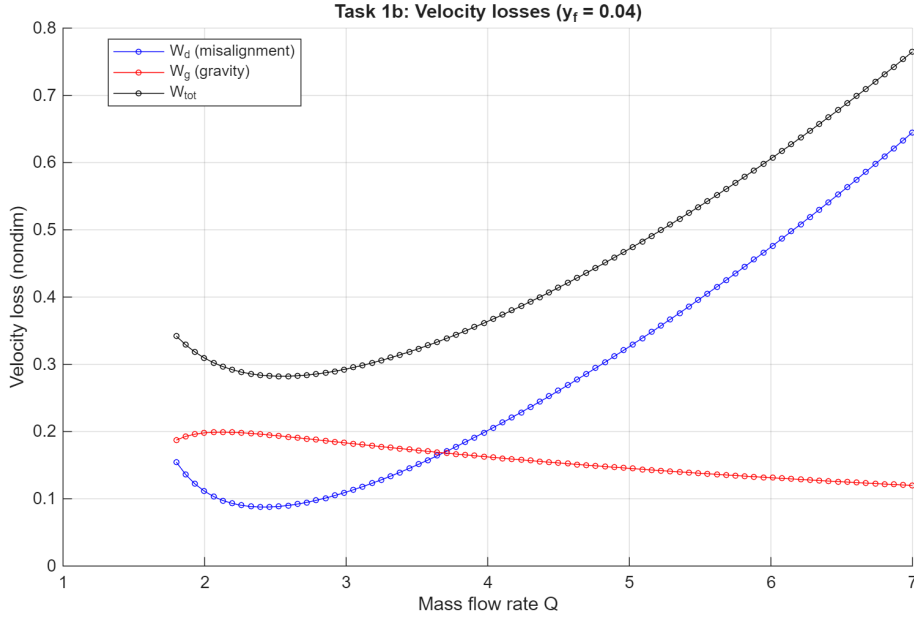
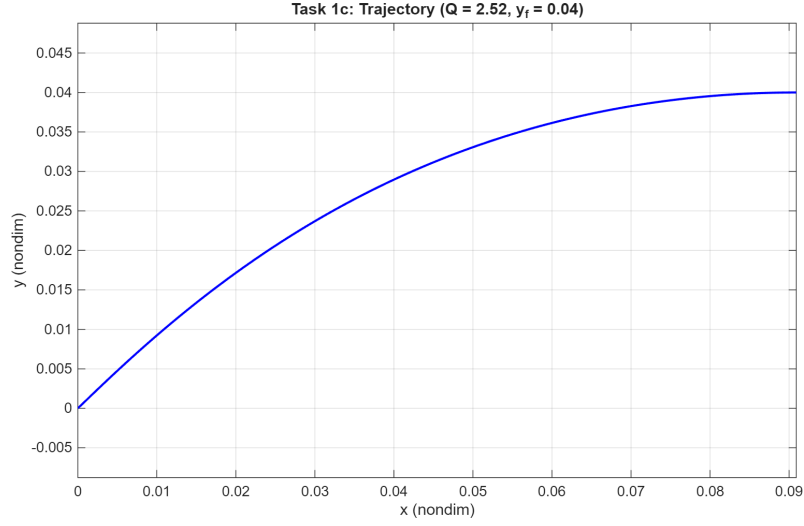
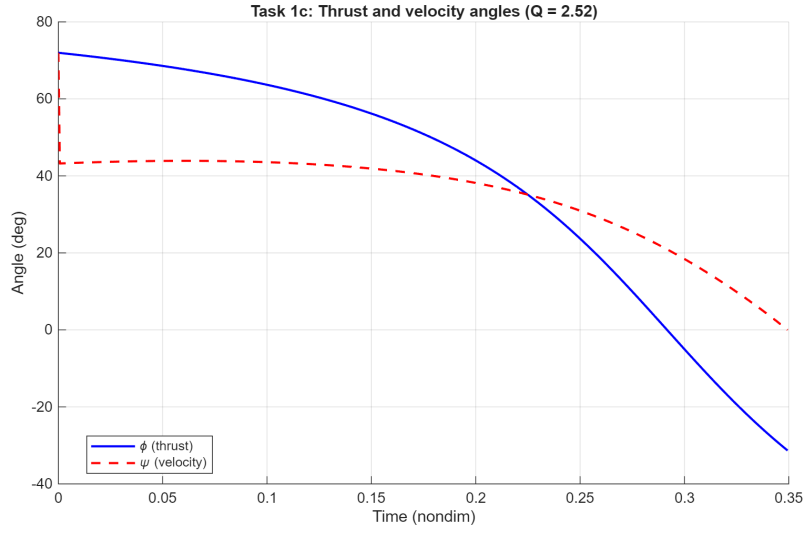


Figure 4: Task 1b: velocity-loss decomposition—misalignment  $W_d$ , gravity  $W_g$ , and total  $W_{\text{tot}}$ —at  $y_f = 0.04$ .

Figures 5a and 5b show the optimal trajectory and control history at  $Q = Q^*$  for  $y_f = 0.04$ . The thrust angle  $\varphi$  decreases smoothly from near-vertical at launch to near-horizontal at injection, following the bilinear-tangent law (16). The velocity angle  $\psi$  tracks  $\varphi$  closely once the initial transient has died out.



(a) Optimal trajectory in the  $(x, y)$  plane.



(b) Thrust angle  $\varphi$  and velocity angle  $\psi$  vs. time.

Figure 5: Task 1c: optimal trajectory and control at  $Q = Q^*$ ,  $y_f = 0.04$ .

## 3 Task 2: Vertical Climb and Optimal Burn

### 3.1 Phase 1: Vertical Climb

Real launchers cannot pitch over right at liftoff, both because of aerodynamic loads and because they have to clear the tower first. Task 2 adds a short vertical-climb phase in front of the burn ( $\varphi = \pi/2$ , thrust pointing straight up) until the altitude reaches  $y_1 = 10^{-4}$  (dimensionally  $\approx 620$  m). During this phase the horizontal position and velocity remain zero, and the 3-state sub-problem is

$$\dot{y} = v_y, \quad \dot{v}_y = \frac{T}{m} - 1, \quad \dot{m} = -Q, \quad (35)$$

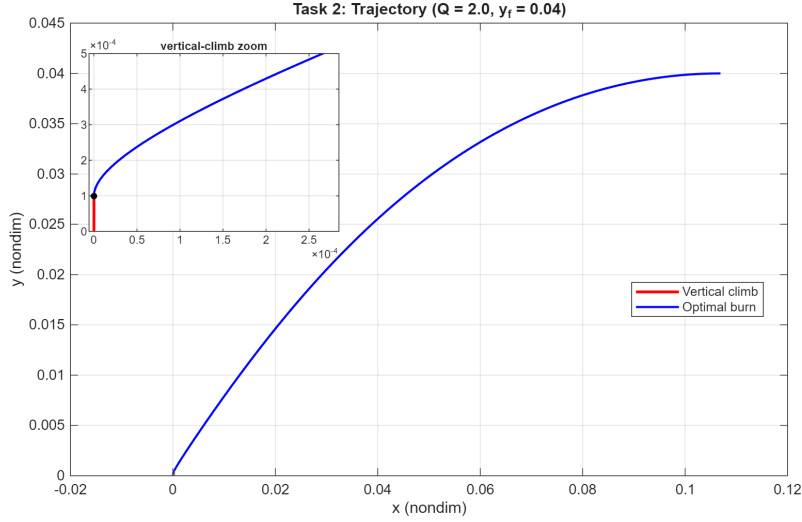
integrated until the event  $y = y_1$  is detected via `ode45` event handling. The end state  $(y_1, v_{y,1}, m_1)$  becomes the initial condition for Phase 2.

### 3.2 Phase 2: Optimal Burn

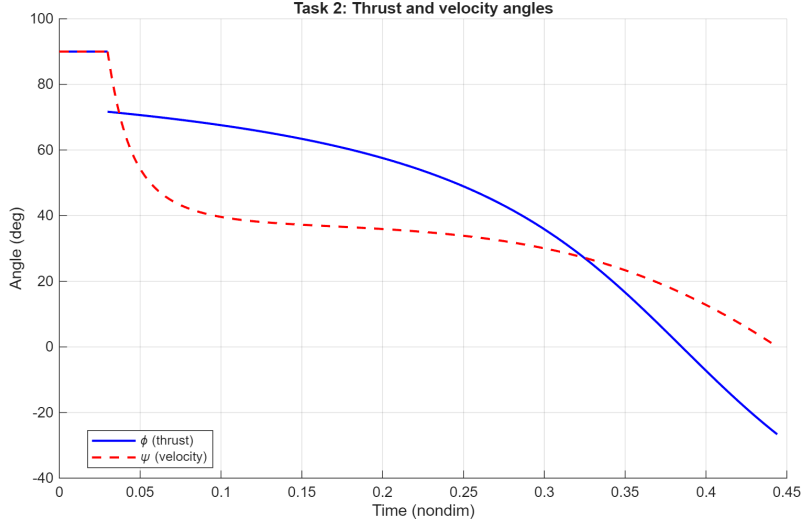
Starting from the state  $(x_0, y_0, v_{x,0}, v_{y,0}, m_0) = (0, y_1, 0, v_{y,1}, m_1)$ , Phase 2 is an optimal powered arc with the same BVP structure as Task 1. The shooting unknowns and the residual (29) are unchanged; only the initial conditions of the augmented state differ. Because the orbit injection targets are identical to Task 1, the continuation warm-start from Task 1's solution at  $Q = 2$  provides a reliable initial guess.

### 3.3 Results

Figure 6 shows the composite trajectory and the angle history for  $Q = 2$ ,  $y_f = 0.04$ . The red segment is the short vertical-climb phase; the blue arc is the optimal burn. The trajectory is qualitatively similar to Task 1 except for the initial vertical stub. During the climb both  $\varphi$  and  $\psi$  are fixed at  $90^\circ$ ; at the transition the thrust begins to rotate toward the horizontal following the bilinear-tangent law, while the velocity angle  $\psi$  lags momentarily before catching up as horizontal speed builds.



(a) Composite trajectory: vertical climb (red), optimal burn (blue).



(b) Thrust angle  $\varphi$  and velocity angle  $\psi$  during the burn.

Figure 6: Task 2: trajectory and control history for  $Q = 2$ ,  $y_f = 0.04$ .

The final mass in Task 2 is marginally lower than in Task 1: the vertical climb consumes propellant while building only vertical velocity, contributing to gravity losses without advancing the vehicle toward the horizontal injection speed. Nevertheless, the prepended climb is a standard launch practice, and the payload penalty is modest for the short  $y_1$  value used here. The converged simulation results are collected in Table 4.



Table 4: Task 2 simulation results ( $Q = 2$ ,  $y_f = 0.04$ ). All quantities are non-dimensional; the Task 1 final mass at the same  $Q$  is reported for reference.

| Quantity                       | Symbol            | Value  |
|--------------------------------|-------------------|--------|
| Vertical-climb duration        | $t_1$             | 0.0298 |
| Powered-burn duration          | $t_{\text{burn}}$ | 0.4142 |
| Total flight time              | $t_f$             | 0.4440 |
| Final mass                     | $m_f$             | 0.1119 |
| Delivered payload              | $m_u$             | 0.0231 |
| Final mass — Task 1 (no climb) | $m_f^{(1)}$       | 0.1128 |

## 4 Task 3: Vertical Climb, Burn, and Coast

### 4.1 Switching Function and Thrust-Arc Structure

Task 3 introduces a coast arc between the powered burn and orbital injection. Unlike Tasks 1 and 2, where the engine burns throughout at the fixed magnitude  $T = cQ$ , the engine may now shut down: the thrust magnitude becomes a bounded control  $T \in [0, T_{\max}]$  (with  $T_{\max} = cQ$ ), alongside the steering angle  $\varphi$ , and the mass flow is  $\dot{m} = -T/c$ . Collecting in the Hamiltonian the terms proportional to  $T$  gives

$$\mathcal{H} = \underbrace{\lambda_x v_x + \lambda_y v_y - \lambda_{v_y}}_{\mathcal{H}_0 \text{ (independent of } T)} + T \left( \frac{\lambda_{v_x} \cos \varphi + \lambda_{v_y} \sin \varphi}{m} - \frac{\lambda_m}{c} \right), \quad (36)$$

where the mass term  $-\lambda_m \dot{m} = -\lambda_m T/c$  now sits inside the  $T$ -bracket because  $T$  is no longer constant. Inserting the optimal steering of the bilinear-tangent law, which maximizes the bracket by aligning the thrust with the velocity costate so that  $\lambda_{v_x} \cos \varphi + \lambda_{v_y} \sin \varphi = \|\boldsymbol{\lambda}_v\|$  (Eq. 19), the Hamiltonian becomes linear in the thrust magnitude:

$$\mathcal{H} = \mathcal{H}_0 + T S(t), \quad S(t) = \frac{\|\boldsymbol{\lambda}_v(t)\|}{m(t)} - \frac{\lambda_m(t)}{c}. \quad (37)$$

The coefficient  $S(t)$  of  $T$  is the **switching function** [4, Sec. 3.9]: the marginal change of the Hamiltonian per unit thrust, i.e. the propulsive benefit  $\|\boldsymbol{\lambda}_v\|/m$  (alignment of thrust with the primer vector  $\boldsymbol{\lambda}_v$  [1, Lecture 6]) net of the propellant penalty  $\lambda_m/c$ . By Pontryagin's maximum principle the optimal  $T$  maximizes  $\mathcal{H}$  pointwise over  $[0, T_{\max}]$ ; since  $\mathcal{H}$  is linear in  $T$ , the maximizer lies at a bound—the **bang–bang** law

$$T^* = \begin{cases} T_{\max}, & S > 0 \quad (\text{engine on}), \\ 0, & S < 0 \quad (\text{coast}). \end{cases} \quad (38)$$

A coast arc is therefore an interval where  $S < 0$ ; there  $T = 0$ , so  $\dot{m} = 0$  and the vehicle flies ballistically. Because  $\dot{\lambda}_m = 0$  whenever  $T = 0$  and the transversality condition (24) fixes  $\lambda_m(t_f) = 1$ , the mass costate stays constant at  $\lambda_m \equiv 1$  along the coast; by continuity  $\lambda_m(t_c) = 1$  at cutoff, so there the switching function reduces to  $S(t_c) = \|\boldsymbol{\lambda}_v(t_c)\|/m(t_c) - 1/c$ .

**Non-singularity of the coast switch.** The bang–bang law assumes that  $S(t)$  vanishes only at isolated instants. If instead  $S \equiv 0$  on a finite interval, the thrust would be singular, left undetermined by the first-order condition. That does not happen here: along the powered arc  $\boldsymbol{\lambda}_v$  is affine in time,  $m$  is positive and linear, and  $\lambda_m$  is monotone (Eq. 14), so  $S = \|\boldsymbol{\lambda}_v\|/m - \lambda_m/c$  is real-analytic in time and can vanish only at isolated points. The thrust profile is therefore bang–bang: a single switch from  $T_{\max}$  to 0 at engine cutoff, with no singular sub-arc.

The three-phase mission sequence is:

1. **Vertical climb:**  $\varphi = \pi/2$  until  $y = y_1 = 10^{-4}$  (identical to Task 2),
2. **Powered burn:** bilinear-tangent steering until engine cutoff at time  $t_c$ ,
3. **Coast arc:** ballistic flight from  $t_c$  to the injection point.

## 4.2 Engine Cutoff Conditions

The shooting unknowns remain  $\boldsymbol{\zeta} = [\lambda_{v_x,0}, \lambda_{v_y,0}, \lambda_y, \lambda_m,0, t_{\text{burn}}]^T$  (with time measured from the start of Phase 2). At engine cutoff the following five conditions must be satisfied simultaneously:

$$\mathbf{r}_3(\boldsymbol{\zeta}) = \begin{bmatrix} v_{x,c} - 1 \\ y_c + \frac{1}{2}v_{y,c}^2 - y_f \\ \lambda_m(t_c) - 1 \\ \lambda_{v_y}(t_c) - \lambda_y v_{y,c} \\ S(t_c) \end{bmatrix} = \mathbf{0}. \quad (39)$$

Conditions 1 and 2 are the terminal constraints  $v_x(t_f) = 1$  and  $y(t_f) = y_f$  carried back to the cutoff through the ballistic coast. The coast leaves  $v_x$  unchanged, so the orbital horizontal speed must already be reached at cutoff:  $v_{x,c} = 1$  (condition 1). The vertical motion is a parabola under constant gravity ( $\dot{v}_y = -1$ ), so the coast reaches apogee ( $v_y = 0$ ) at  $\tau^* = v_{y,c}$  with altitude  $y_c + \frac{1}{2}v_{y,c}^2$  (Section 4.3); imposing  $y(t_f) = y_f$  gives condition 2. The same result is vertical energy conservation along the coast: with  $g = 1$ ,  $\frac{1}{2}v_y^2 + y$  is constant, so  $\frac{1}{2}v_{y,c}^2 + y_c = y_f$  — the residual vertical kinetic energy at cutoff is exactly what lifts the vehicle from  $y_c$  to the target altitude. Condition 3 is the standard mass-costate transversality (24). Condition 4 is the free-final-time condition  $\mathcal{H} = 0$  (25), read on the coast. With  $T = 0$  and  $\lambda_x \equiv 0$  the coast Hamiltonian reduces to  $\mathcal{H} = \lambda_y v_y - \lambda_{v_y}$ , which is constant and equal to zero along the optimal trajectory (the same  $\mathcal{H} \equiv 0$  I impose at  $t_0$  in Task 1); evaluating it at cutoff, where  $v_y = v_{y,c}$ , gives  $\lambda_{v_y}(t_c) = \lambda_y v_{y,c}$  directly. Equivalently, at apogee ( $v_y = 0$ ) it reads  $\lambda_{v_y}(t_f) = 0$ , which propagates back to the cutoff through the linear evolution  $\lambda_{v_y}(t_f) = \lambda_{v_y}(t_c) - \lambda_y v_{y,c}$ . This is the Hamiltonian (free-time) condition, not the transversality of the constrained terminal  $v_y$ , whose costate is left free. Condition 5 is the bang-bang switching condition: the engine cuts off exactly when  $S = 0$ .

## 4.3 Coast Arc

During the ballistic coast ( $T = 0$ ) the trajectory is analytically integrable. Denoting the elapsed coast time by  $\tau$  and the cutoff state by subscript  $c$ :

$$x(t_c + \tau) = x_c + v_{x,c} \tau, \quad y(t_c + \tau) = y_c + v_{y,c} \tau - \frac{1}{2}\tau^2. \quad (40)$$

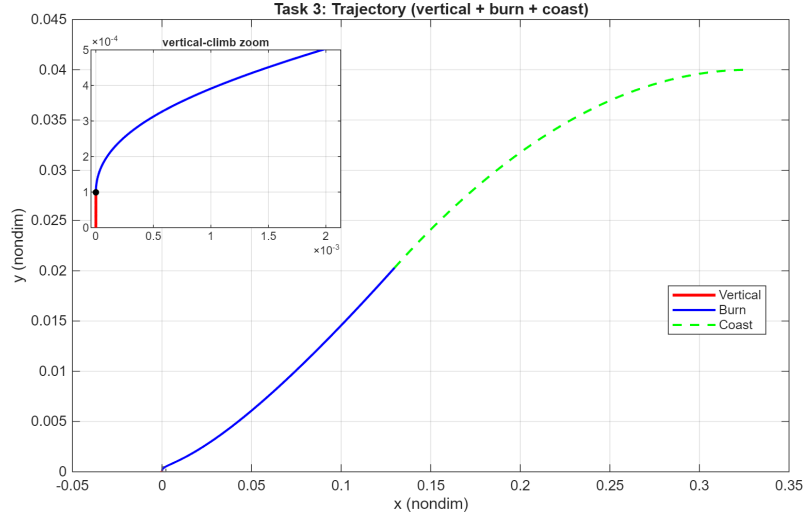
The coast ends at apogee  $\tau^* = v_{y,c}$ , where  $v_y = 0$  and

$$y_f = y_c + \frac{1}{2}v_{y,c}^2, \quad (41)$$

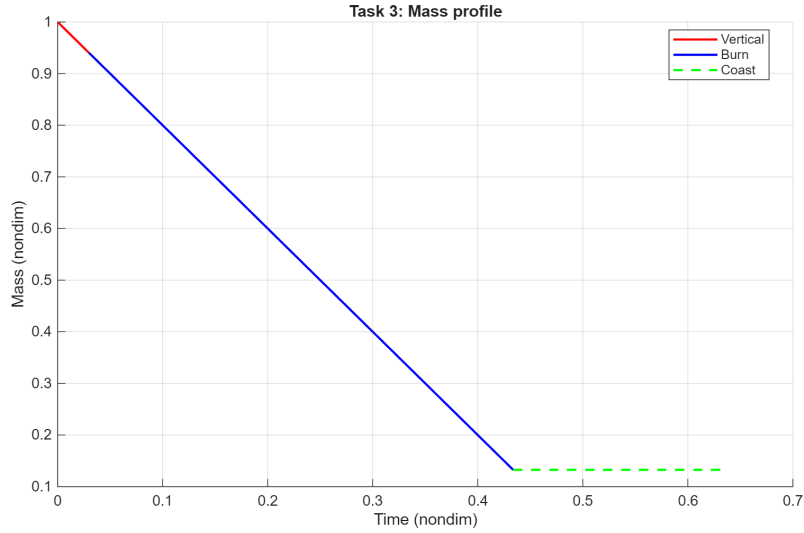
in agreement with condition 2 of (39). The vehicle mass is constant throughout the coast:  $m_f = m_c$ .

## 4.4 Results

Figure 7 shows the three-phase trajectory and mass profile for  $Q = 2$ ,  $y_f = 0.04$ . The vertical climb (red) is short; the burn arc (blue) terminates below the target altitude; the ballistic coast (green dashed) ascends to the injection point. The mass profile confirms the model: it decreases linearly during both powered phases and remains constant during the coast. The final mass in Task 3 exceeds that of Task 2: by cutting the engine early and relying on the ballistic coast, less propellant is consumed while still satisfying all terminal constraints.



(a) Trajectory: vertical climb (red), burn (blue), coast (green dashed).



(b) Mass profile: linear decrease during powered phases, flat during coast.

Figure 7: Task 3: trajectory and mass profile for  $Q = 2$ ,  $y_f = 0.04$ .

Figure 8 shows the thrust and velocity angles throughout the three phases. The dashed vertical lines mark the end of the vertical climb and the engine cutoff. During the coast no thrust angle is defined; the velocity angle  $\psi$  continues to evolve ballistically until it reaches zero at apogee.

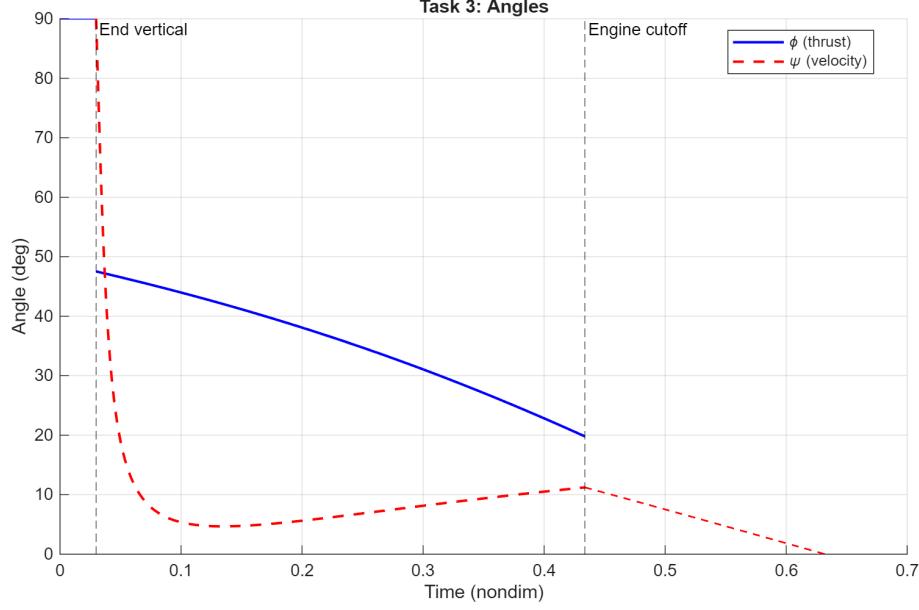


Figure 8: Task 3: thrust angle  $\varphi$  and velocity angle  $\psi$  across all three phases. Vertical dashed lines mark phase transitions.

A clear difference from Task 2 shows up in the thrust angle just after the vertical climb:  $\varphi \approx 47^\circ$  here, against  $\approx 71^\circ$  in Task 2.<sup>2</sup> Because the coast covers the upper half of the climb (from  $y_c \approx 0.020$  up to  $y_f = 0.040$ ) with the engine off, the burn no longer has to gain all the altitude on its own and can point the thrust closer to the horizontal from the start, that is, a more aggressive gravity turn that spends more of the thrust on horizontal acceleration. The same logic appears at cutoff: in Task 2 the thrust angle becomes negative ( $\varphi \approx -27^\circ$ ), with the engine pushing downward in the last instants to null  $v_y$  at injection, while here it stays positive ( $\varphi \approx +20^\circ$ ) and the engine simply shuts off while  $v_y$  is still positive ( $v_{y,c} = 0.199$ ), letting gravity null it during the coast. Cutting off near-horizontally with some vertical velocity left over, instead of burning propellant to brake vertically, is what raises the final mass from  $m_f = 0.112$  (Task 2) to 0.133.

The velocity angle  $\psi$  shows the same effect on the trajectory side. In Task 2 it falls from  $90^\circ$  to a plateau near  $37^\circ$  and then decreases monotonically to  $0^\circ$  at injection, with the engine still firing. In Task 3 it instead drops to a much lower minimum ( $\approx 5^\circ$ ), then rises back to  $\approx 11^\circ$  at cutoff, and only reaches  $0^\circ$  at apogee, with the engine off. The deep minimum is the counterpart of the smaller thrust angle: with  $\varphi \approx 47^\circ$  and the mass still high just after the climb, the vertical thrust component  $\frac{T}{m} \sin \varphi$  is smaller than  $g$ , so  $v_y$  drops and the path flattens sharply; as propellant burns off,  $\frac{T}{m}$  grows until  $\frac{T}{m} \sin \varphi > g$  again,  $v_y$  climbs back and  $\psi$  recovers, which is how the

<sup>2</sup>The thrust-angle, flight-path-angle and velocity values quoted in this paragraph ( $\varphi$ ,  $\psi$ ,  $v_{y,c}$ ) are read from the converged solutions computed by `main_task2.m` and `main_task3.m`, rather than measured off the figures.

burn builds up the residual  $v_{y,c} = 0.199$  handed to the coast. In Task 2 the larger  $\varphi$  keeps  $\frac{T}{m} \sin \varphi > g$  from the start, so  $v_y$  never collapses,  $\psi$  holds its plateau, and the engine itself drives  $\psi$  to zero. The U-shaped  $\psi$  in Task 3, compared with the monotone  $\psi$  in Task 2, is the signature of the coast on the trajectory.

## 5 Task 4: Optimal Staging

### 5.1 Staging Model

Task 4 removes the vertical-climb phase and introduces a single staging event at time  $t = t_s$ . At staging, the first-stage structural mass

$$m_{s1} = \eta Q t_s \quad (42)$$

is jettisoned instantaneously. Here  $Q t_s$  is the propellant the first stage burns up to  $t_s$ , and  $\eta = m_s/m_p$  is the structural coefficient (inert mass per unit propellant,  $\eta = 0.1$ ): to hold that propellant the stage has to carry a tankage/engine mass  $\eta Q t_s$ . The propellant is fully consumed—the mass falls as  $m(t) = 1 - Q t$ —and only the inert structure is dropped, so the vehicle mass undergoes a step:

$$m^+(t_s) = m^-(t_s) - m_{s1} = 1 - Q t_s - \eta Q t_s = 1 - Q t_s(1 + \eta). \quad (43)$$

For  $m^+(t_s) > 0$  one requires  $t_s < [Q(1 + \eta)]^{-1}$ , which provides an upper bound on the admissible staging time. The position and velocity states are continuous across the jettison—only the mass steps—and, being unconstrained interior values, the PMP interior corner conditions keep their costates  $\lambda_x, \lambda_y, \lambda_{v_x}, \lambda_{v_y}$  continuous across  $t_s$  as well [1, Lecture 5]. The bilinear-tangent steering law is therefore unaffected by staging.

The net payload delivered to orbit is

$$m_u = m_f - m_{s2}, \quad m_{s2} = \eta Q (t_f - t_s), \quad (44)$$

where  $m_{s2}$  is the second-stage structural mass (again  $\eta$  times the propellant that stage burns). Summing the budget over both stages—a total propellant  $Q t_f$ , its structure  $\eta Q t_f$ , and the payload, all drawn from  $m_0 = 1$ —collapses to the compact relation

$$m_u = 1 - (1 + \eta) Q t_f, \quad (45)$$

which depends only on the total burn time  $t_f$  (equivalently, the total propellant). Staging raises the payload because dropping the spent first-stage structure lets the vehicle reach orbit on a smaller total propellant load, i.e. a smaller  $t_f$ : here  $t_f$  drops from 0.444 (single stage) to 0.424 (two-stage), lifting  $m_u$  from 0.024 to 0.068.

### 5.2 Optimization Procedure

For a fixed staging time  $t_s$ , the two burn arcs share the same costate evolution (the bilinear-tangent law is unaffected by staging), and the BVP reduces to the same structure as Task 1 with updated initial mass after staging:

1. Integrate the augmented state from  $t = 0$  to  $t = t_s$  (Stage 1 burn),



2. Apply the staging mass jump (42),
3. Integrate from  $t_s$  to  $t_f$  (Stage 2 burn),
4. Evaluate the residual (29) at  $t_f$ .

The shooting unknowns  $\zeta$  (Eq. 27) are identical to Task 1.

The optimal staging time  $t_s^*$  is found by an **outer parametric sweep** (implemented in `main.task4.m`):  $t_s$  is varied over a dense grid of 50 points, the inner BVP is solved at each point (warm-started from the previous converged solution via continuation, as in Listing 3), and the payload (44) is maximized over the converged solutions. The starting guess for the inner BVP at the first grid point uses the single-stage solution from Task 1.

A fully indirect treatment closes the free interior time  $t_s$  with a corner condition at the jettison rather than sweeping for it. Since the delivered payload (45) depends only on the total burn time  $t_f$ , maximising it is the same as minimising  $t_f$ , and the optimum is the stationary point

$$\frac{dt_f}{dt_s} = 0, \quad (46)$$

the peak of the  $m_u(t_s)$  curve (Figure 9). Staging is a three-point junction at the free time  $t_s$ , so the interior-point conditions of Bryson [4, Sec. 3.5] govern it: since the jettisoned mass depends on  $t_s$  alone and not on the state, the position, velocity and mass costates all stay continuous — and with them the bilinear-tangent steering — while only the Hamiltonian jumps. Equating its jump from the thrust term  $(T/m)\|\lambda_v\|$  to its jump from the explicit  $t_s$  of the jettisoned mass  $m_{s1} = \eta Q t_s$ , carried out step by step in Appendix C, gives the corner condition

$$\begin{aligned} \eta[\lambda_m(t_f) - \lambda_m(t_s)] &= c \|\lambda_v(t_s)\| \left( \frac{1}{m^+} - \frac{1}{m^-} \right), \\ m^- &= 1 - Q t_s, \quad m^+ = 1 - (1 + \eta) Q t_s, \end{aligned} \quad (47)$$

which balances the inert mass shed at staging (left) against the propulsive gain of the lighter vehicle over the rest of the burn (right). The mass costate enters in the burnout-referenced combination  $\lambda_m(t_f) - \lambda_m(t_s)$  because the staging time minimises the total burn time: the final mass is free, so transversality fixes  $\lambda_m(t_f) = 0$ , and taking the difference makes the condition independent of the inner BVP's  $\lambda_{m,0} = 1$  scaling. Appending  $t_s$  as a fifth shooting unknown and (47) as a fifth residual returns  $t_s^* \approx 0.336$ , matching the swept grid value of Table 5; the outer sweep is an equivalent but more transparent route to the same optimum.

### 5.3 Results

Figure 9 shows the payload as a function of staging time  $t_s$  for  $Q = 2$ ,  $y_f = 0.04$ . The curve has a well-defined maximum, and the single-stage baseline (dashed red) lies well

below the two-stage peak.

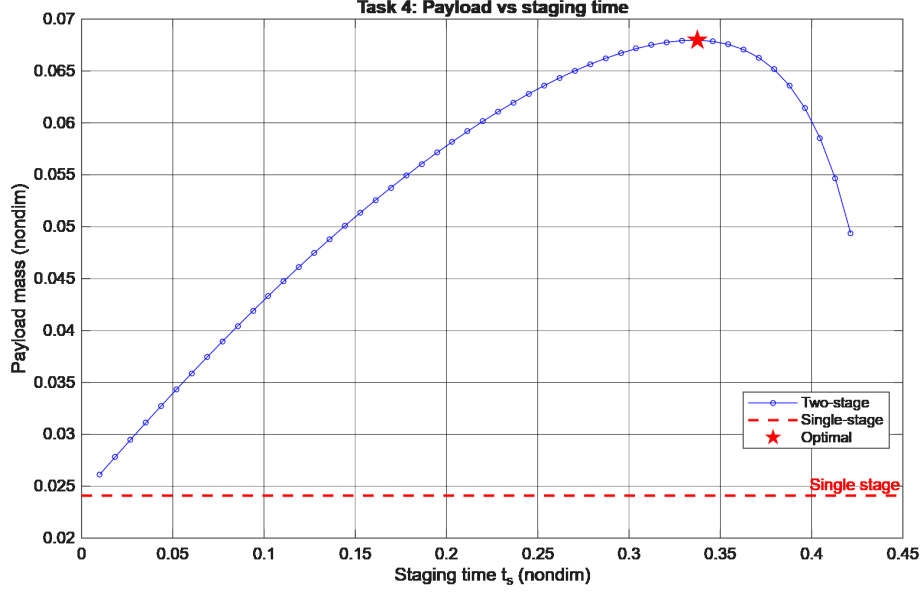


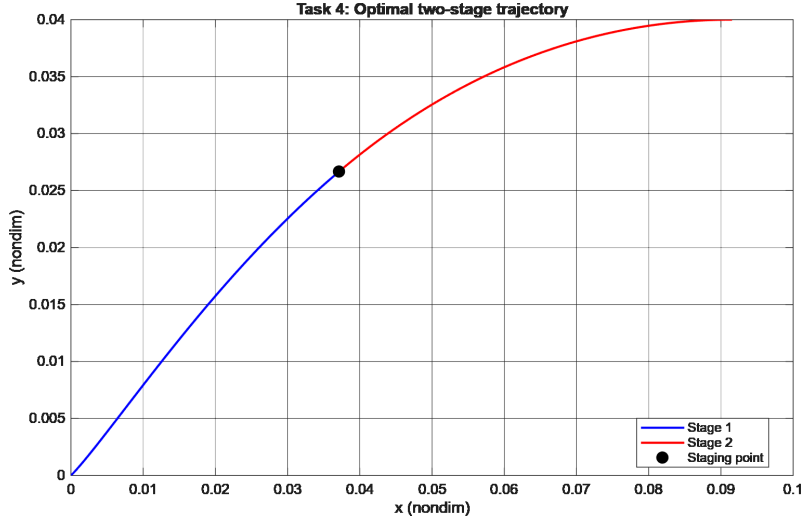
Figure 9: Task 4: payload mass vs. staging time  $t_s$ . Red dashed line: single-stage reference. Red star: optimal staging point.

Table 5 summarises the numerical results at  $Q = 2$ ,  $y_f = 0.04$ .

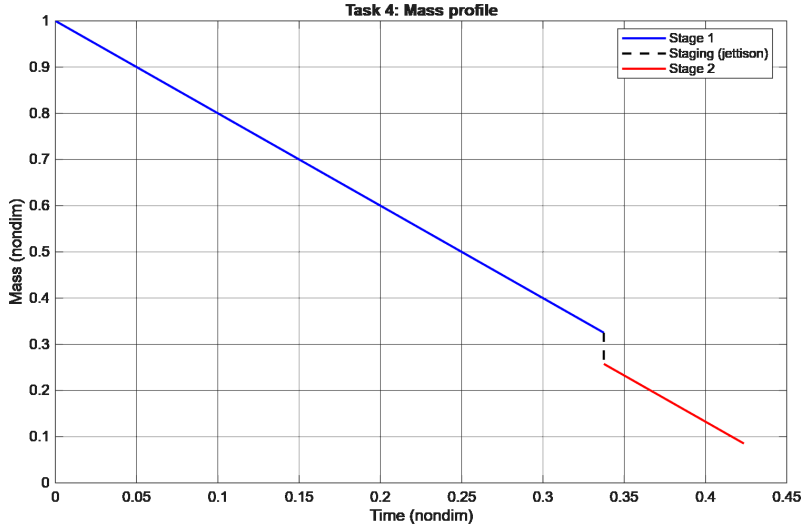
Table 5: Task 4 key numerical results ( $Q = 2$ ,  $y_f = 0.04$ ).

| Quantity                     | Value  | Notes                     |
|------------------------------|--------|---------------------------|
| Optimal staging time $t_s^*$ | 0.337  | nondim ( $\approx 268$ s) |
| Single-stage payload         | 0.0241 | nondim                    |
| Two-stage payload            | 0.0680 | nondim                    |
| Payload gain                 | +182%  | vs. single-stage          |

Figures 10a and 10b show the optimal two-stage trajectory and mass profile. The discontinuous mass drop at  $t_s^*$  is visible in the profile; the trajectory is continuous (position and velocity are unaffected by staging, only mass changes).



(a) Optimal two-stage trajectory: Stage 1 (blue), Stage 2 (red), staging point (circle).



(b) Mass profile with the instantaneous jettison at  $t_s^*$ .

Figure 10: Task 4: optimal two-stage solution for  $Q = 2$ ,  $y_f = 0.04$ .

The large relative gain is a consequence of the model simplifications (constant  $Q$ , no drag, identical  $I_{sp}$  for both stages). Real launchers achieve much smaller but still substantial staging benefits; the key mechanism—jettisoning inert structural mass before it has to be accelerated to orbital velocity—is the same.

## 6 Conclusions

This work applied the Pontryagin Maximum Principle to four minimum-fuel planar ascent programs of increasing complexity, all solved by single-shooting with `fsolve` over the costate initial conditions. Table 6 summarises the four programs at the common operating point  $Q = 2$ ,  $y_f = 0.04$ ; the main findings are discussed below.

Table 6: Comparison of the four ascent programs at  $Q = 2$ ,  $y_f = 0.04$  (non-dimensional). The payload of the single-stage programs (Tasks 1–3) is  $m_u = m_f(1 + \eta) - \eta$ ; for the two-stage program (Task 4) it is the net mass after jettisoning both stage structures.

| Program | Mission phases              | $t_f$ | $m_f$  | $m_u$  |
|---------|-----------------------------|-------|--------|--------|
| Task 1  | burn                        | 0.444 | 0.1128 | 0.0241 |
| Task 2  | vertical + burn             | 0.444 | 0.1119 | 0.0231 |
| Task 3  | vertical + burn + coast     | 0.632 | 0.1326 | 0.0458 |
| Task 4  | stage 1 + stage 2 (staging) | 0.424 | 0.0852 | 0.0680 |

The delivered payload goes  $0.0241 \rightarrow 0.0231 \rightarrow 0.0458 \rightarrow 0.0680$  across the four programs, which shows the trade-offs at work: the vertical climb (Task 2) costs a few percent of payload compared with the bare burn (Task 1), the ballistic coast (Task 3) nearly doubles it (+90% over Task 1), and optimal staging (Task 4) almost triples it (+182%). The main findings are summarized below.

**Bilinear-tangent law (all tasks).** The flat-Earth, drag-free dynamics produce a particularly tractable adjoint structure: four of the five costate components are determined analytically (two constants and one linear ramp), leaving only the mass costate  $\lambda_m$  to be integrated numerically. The resulting bilinear-tangent steering law reduces the optimal control problem to a four-dimensional root-finding problem for the powered-arc programs (Tasks 1, 2 and 4) and a five-dimensional one for Task 3, where the mass costate is retained to evaluate the coast switching condition—making single-shooting computationally affordable in either case.

**Optimal thrust-to-weight ratio (Task 1).** The mass-flow rate  $Q$  that maximises the final mass balances two competing losses: gravity losses (dominant at low  $Q$ , due to long burn times) and thrust-misalignment losses (dominant at high  $Q$ , due to steep climb angles). The optimal  $Q^*$  shifts slightly with target altitude  $y_f$ , reflecting the longer burn time required for higher orbits.

**Continuation is essential.** Single shooting from a cold start fails across most of the  $Q$  range. The continuation strategy—warm-starting each new  $Q$  from the previous

converged solution and sweeping forward and backward from a seed point—was the key ingredient that made the parameter sweeps in Tasks 1 and 4 possible.

**Coast arc saves propellant (Task 3).** Terminating the engine before reaching the target altitude and relying on ballistic dynamics is beneficial whenever the vehicle has sufficient residual vertical velocity to coast up to the injection point. The condition  $S(t_c) = 0$  determines the optimal engine cutoff time without an additional outer optimization loop.

**Staging provides large payload gains (Task 4).** At the reference operating point ( $Q = 2$ ,  $y_f = 0.04$ ), the optimal two-stage configuration delivers a +182% payload gain over the single-stage baseline. Even the simplified constant- $Q$ , drag-free model already captures most of this gain.

**Model limitations.** The simplifications used throughout (flat Earth, no drag, constant thrust, no vehicle attitude dynamics) leave out important physical effects. Atmospheric drag is large in the dense lower atmosphere—its density falls off with an  $\approx 7\text{--}8$  km scale height in the standard exponential model, and capturing it needs an ascent benchmark that includes drag [3, Sec. 6.15]; Earth’s curvature and rotation change the optimal coast and staging conditions; and real throttling constraints keep  $Q$  away from its full theoretical range. Adding a drag model would shift  $Q^*$  and  $t_s^*$  in value and, more importantly, would change the whole structure of the indirect problem. Because drag depends on altitude (through  $\rho$ ) and on speed (through  $V$ ), the costates  $\lambda_y, \lambda_{v_x}, \lambda_{v_y}$  would no longer be constant or linear: the closed-form adjoints and the bilinear-tangent law would be lost, and all five costates would need numerical integration with a stiffer shooting problem. The optimal thrust direction would still follow the primer vector ( $\tan \varphi = \lambda_{v_y}/\lambda_{v_x}$ , since drag does not depend on  $\varphi$ ), and the PMP plus single-shooting machinery would still apply, but the analytic structure that makes the present drag-free problem so easy to solve is a direct consequence of neglecting aerodynamics.

## A A Canonical Indirect/Shooting Benchmark: the Rocket Sled

Before tackling the ascent BVP, I validated the single-shooting code on a classical minimum-energy problem whose solution is known in closed form, the rocket sled [1, Lecture 5]. A unit point mass moves on a frictionless rail under a force  $u$ ,

$$\dot{r} = v, \quad \dot{v} = u, \quad (48)$$

and must be driven from rest at the origin to rest at unit displacement,

$$r(0) = 0, \quad v(0) = 0, \quad r(2) = \frac{1}{2}, \quad v(2) = 0, \quad (49)$$

in fixed time  $t_f = 2$ , minimizing the control energy  $J = \int_0^2 u^2 dt$ . With the Hamiltonian

$$\mathcal{H} = -u^2 + \lambda_r v + \lambda_v u \quad (50)$$

(written in the same maximization convention used throughout this report), the costate equations are  $\dot{\lambda}_r = 0$  and  $\dot{\lambda}_v = -\lambda_r$ , so  $\lambda_r$  is constant and  $\lambda_v(t) = \lambda_{v,0} - \lambda_r t$  is linear in time. Stationarity  $\partial \mathcal{H} / \partial u = -2u + \lambda_v = 0$  gives the optimal control  $u = \lambda_v / 2$ , also linear in time. Imposing the four boundary conditions fixes the two unknown initial costates analytically,

$$\lambda_{r,0} = \lambda_{v,0} = \frac{3}{2} \quad \implies \quad u^*(t) = \frac{3}{4}(1-t), \quad (51)$$

which yields  $v(t) = \frac{3}{4}(t - t^2/2)$  and  $r(t) = \frac{3}{4}(t^2/2 - t^3/6)$ , satisfying  $r(2) = \frac{1}{2}$  and  $v(2) = 0$  exactly. Running the same `ode45` + `fsolve` shooting machinery on this two-unknown problem (`validate_rocket_sled.m`) recovers  $(\lambda_{r,0}, \lambda_{v,0}) = (\frac{3}{2}, \frac{3}{2})$  to within  $10^{-8}$  and drives the terminal residual to solver tolerance. This checks the implementation before it is applied to the ascent BVP. The rocket sled also shows how fragile shooting can be: its convergence map has sizeable regions of the initial-costate plane from which Newton iterations diverge [1, Lecture 5], which is what motivates the continuation strategy of Section 1.5.

## B Implementation: Key Code Excerpts

The full MATLAB sources (`main_task1.m`–`main_task4.m` and `ode_burn.m`) live in the `HM1/` folder of the repository. Here I collect only the three routines that do the actual work: the powered-flight right-hand side, the shooting residual, and the continuation sweep. The four tasks reuse the same building blocks, so showing them once is enough.

The optimal dynamics are integrated by the right-hand side in Listing 1. Four of the five costates are written in closed form (two constants and the linear ramp  $\lambda_{v_y}$ ), so only  $\lambda_m$  is actually propagated, and the thrust angle is read off the velocity costate with a single `atan2`.

```

1 function dz = ode_burn(t, z, p)
2 % Powered-flight RHS, linear-tangent steering.
3 % INPUT
4 %   t - time (scalar)
5 %   z - state [x; y; vx; vy; m; lam_m]
6 %   p - struct: T, Q, c, lam_vx0, lam_vy0, lam_y
7 % OUTPUT
8 %   dz - state derivative (6x1)
9 %
10 % Costates: lam_x=0, lam_y=p.lam_y, lam_vx=p.lam_vx0,
11 % lam_vy=p.lam_vy0 - p.lam_y*t. Thrust angle phi=atan2(lam_vy,lam_vx).
12 %
13 % No arguments block by design: called ~1e6+ times inside ode45/fsolve;
14 % validate at the call site.
15
16 vx = z(3); vy = z(4); m = z(5);
17
18 % Costates
19 lam_vx = p.lam_vx0;
20 lam_vy = p.lam_vy0 - p.lam_y * t;
21 lam_v_norm = sqrt(lam_vx^2 + lam_vy^2);
22
23 % Optimal thrust angle
24 phi = atan2(lam_vy, lam_vx);
25
26 % State derivatives
27 dz = zeros(6,1);
28 dz(1) = vx; % dx/dt
29 dz(2) = vy; % dy/dt
30 dz(3) = (p.T / m) * cos(phi); % dvx/dt
31 dz(4) = (p.T / m) * sin(phi) - 1; % dvy/dt (g = 1 nondim)
32 dz(5) = -p.Q; % dm/dt
33 dz(6) = (p.T / m^2) * lam_v_norm; % dlam_m/dt
34
35 end

```

Listing 1: Powered-flight right-hand side with linear-tangent steering (`ode_burn.m`).

Listing 2 is the four-residual shooting function shared by Tasks 1, 2 and 4. The two numerical refinements discussed in Section 1.5 are both visible: the state is initialized with  $\lambda_{m,0} = 1$  (the costate normalization), and the free-time condition  $\mathcal{H}(0) = 0$  is imposed algebraically at  $t_0$  rather than at  $t_f$ .

```

227 function res = shooting1(z0, p, opts_ode)
228 % Shooting residual for a single burn arc.
229 % INPUT
230 %     z0      - unknowns [lam_vx0; lam_vy0; lam_y; tf]
231 %     p       - struct: c, T, Q, yf
232 %     opts_ode - ode45 options
233 % OUTPUT
234 %     res - 4 residuals [y(tf)-yf; vx(tf)-1; vy(tf); H0]
235 % Improved formulation: costates normalized to lam_m0=1 (H homogeneous deg 1 in
236 % lambda) and H=0 imposed at t0, where it is algebraic (vx0=vy0=0) and error-free
237 % instead of at tf. 4 unknowns, 4 residuals; lam_m never enters.
238 % No arguments block by design: runs inside the fsolve loop.
239
240 lam_vx0 = z0(1);
241 lam_vy0 = z0(2);
242 lam_y   = z0(3);
243 tf      = z0(4);
244
245 if tf <= 0 || tf > 2
246     res = 1e6 * ones(4,1);
247     return;
248 end
249
250 pp = p;
251 pp.lam_vx0 = lam_vx0;
252 pp.lam_vy0 = lam_vy0;
253 pp.lam_y   = lam_y;
254
255 ic = [0; 0; 0; 0; 1; 1]; % state + lam_m0 = 1 (normalization)
256
257 try
258     [~, Z] = ode45(@(t,z) ode_burn(t, z, pp), [0 tf], ic, opts_ode);
259     zf = Z(end,:);
260 catch
261     res = 1e6 * ones(4,1);
262     return;
263 end
264
265 % H = 0 at t0 (autonomous => H const = 0). With vx0 = vy0 = 0, m0 = 1 and
266 % lam_m0 = 1: H0 = -lam_vy0 + T*( |lam_v0| - 1/c ).
267 Lam0 = sqrt(lam_vx0^2 + lam_vy0^2);
268 H0    = -lam_vy0 + p.T * (Lam0 - 1/p.c);
269
270 res = [zf(2) - p.yf; % y(tf) = yf
271       zf(3) - 1;    % vx(tf) = 1
272       zf(4);        % vy(tf) = 0
273       H0];          % H(0) = 0 (free final time)

```

Listing 2: Shooting residual for a single burn arc (`main_task1.m`).

The parameter sweeps depend on continuation: each new value of  $Q$  is solved starting from the previous converged costate vector, sweeping both forward and backward from a seed point near  $Q = 3$ . Listing 3 shows the two sweeps.

```

65 % Sweep forward (increasing Q)
66 z_prev = z_sol;
67 for ii = idx0+1:nQ
68     p.Q = Q_vec(ii);
69     p.T = c * p.Q;
70     [z_sol, ~, ef] = fsolve(@(z) shooting1(z, p, opts_ode), z_prev, opts_fs);

```



```

71     if ef > 0
72         [~, Z] = ode45(@(t,z) ode_burn(t,z,set_costates(p,z_sol)), ...
73             [0 z_sol(4)], [0;0;0;0;1;1], opts_ode);
74         mf_results(ii,jj) = Z(end,5);
75         sol_results{ii,jj} = z_sol;
76         z_prev = z_sol;
77     end
78 end
79
80 % Sweep backward (decreasing Q)
81 z_prev = sol_results{idx0,jj};
82 for ii = idx0-1:-1:1
83     p.Q = Q_vec(ii);
84     p.T = c * p.Q;
85     [z_sol, ~, ef] = fsolve(@(z) shooting1(z, p, opts_ode), z_prev, opts_fs);
86     if ef > 0
87         [~, Z] = ode45(@(t,z) ode_burn(t,z,set_costates(p,z_sol)), ...
88             [0 z_sol(4)], [0;0;0;0;1;1], opts_ode);
89         mf_results(ii,jj) = Z(end,5);
90         sol_results{ii,jj} = z_sol;
91         z_prev = z_sol;
92     end
93 end

```

Listing 3: Continuation sweep in  $Q$  (excerpt from the per-altitude loop, `main_task1.m`).

## C Step-by-step derivation of the staging corner condition

This appendix derives the optimality condition for the free staging time  $t_s$  of Task 4 (Eq. (47)) one step at a time, in the costates the shooting BVP actually integrates (normalized by  $\lambda_{m,0} = 1$ ). Every step carries an explicit sign check.

**Step 1 — the objective is the total burn time.** The delivered payload (45),  $m_u = 1 - (1 + \eta)Qt_f$ , depends only on the total burn time  $t_f$ . Maximizing  $m_u$  is therefore identical to minimizing  $t_f$ , and the interior optimum in  $t_s$  is the stationary point  $dt_f/dt_s = 0$  (46). The rest of the appendix rewrites this stationarity as a condition on the costates at the jettison.

**Step 2 — costates on the two arcs.** On each powered arc, with the bilinear-tangent steering inserted, the Hamiltonian is (20)

$$\mathcal{H} = \lambda_y v_y + \frac{T}{m} \|\boldsymbol{\lambda}_v\| - \lambda_{v_y} - \lambda_m Q, \quad \lambda_x \equiv 0. \quad (52)$$

The adjoint equations give  $\lambda_{v_x}$  and  $\lambda_y$  constant,  $\lambda_{v_y}$  affine in time, and  $\dot{\lambda}_m = (T/m^2)\|\boldsymbol{\lambda}_v\| > 0$  (14). This last point is used repeatedly:  $\lambda_m$  is strictly increasing, so  $\lambda_m(t_s) < \lambda_m(t_f)$ .

**Step 3 — the junction as an interior-point condition.** Staging makes the problem a three-point boundary-value problem: arc 1 on  $[0, t_s]$ , arc 2 on  $[t_s, t_f]$ , joined at the free interior time  $t_s$ . With  $\mathbf{z} = (x, y, v_x, v_y, m)$  the two arcs are linked by

$$N \equiv \mathbf{z}(t_s^+) - \mathbf{z}(t_s^-) - \Delta(t_s) = 0, \quad \Delta(t_s) = (0, 0, 0, 0, -\eta Q t_s)^\top : \quad (53)$$

position and velocity carry over unchanged, and the mass drops by the spent first-stage structure  $\eta Q t_s$ . Across an interior junction Bryson's conditions [4, Sec. 3.5] fix the costate and the Hamiltonian in terms of the constant multiplier  $\boldsymbol{\pi}$  of  $N$ ,

$$\boldsymbol{\lambda}^\top(t_s^-) = \boldsymbol{\lambda}^\top(t_s^+) + \boldsymbol{\pi}^\top N_{\mathbf{z}}, \quad \mathcal{H}(t_s^-) = \mathcal{H}(t_s^+) - \boldsymbol{\pi}^\top N_t, \quad (54)$$

which are Bryson's Eqs. (3.5.4) and (3.5.5).

The junction (53) couples the two arcs only through  $N$ , which therefore carries one state block per side:

$$N_{\mathbf{z}^-} \equiv \frac{\partial N}{\partial \mathbf{z}(t_s^-)} = -I_5, \quad N_{\mathbf{z}^+} \equiv \frac{\partial N}{\partial \mathbf{z}(t_s^+)} = +I_5, \quad (55)$$

exact identities with no off-diagonal or state-dependent entries, because the jettisoned mass  $\Delta(t_s)$  is a function of  $t_s$  alone ( $\partial \Delta / \partial \mathbf{z} = 0$ ).

Conditions (54) are written for a one-sided junction, where the state is continuous and there is a single block  $N_x$ ; here the state jumps, so the linkage ties the two distinct values  $\mathbf{z}(t_s^-)$  and  $\mathbf{z}(t_s^+)$ , and the variational argument behind (54) is worth carrying out in full. Adjoin the linkage to the cost with multiplier  $\boldsymbol{\pi}$  and append the dynamics  $\dot{\mathbf{z}} = \mathbf{f}$  on each arc through the costate:

$$\begin{aligned} \bar{J} = & \phi(\mathbf{z}(t_f)) + \boldsymbol{\pi}^\top N \\ & + \int_0^{t_s} \boldsymbol{\lambda}^\top (\mathbf{f} - \dot{\mathbf{z}}) dt + \int_{t_s}^{t_f} \boldsymbol{\lambda}^\top (\mathbf{f} - \dot{\mathbf{z}}) dt. \end{aligned} \quad (56)$$

Integrating the  $-\boldsymbol{\lambda}^\top \dot{\mathbf{z}}$  term by parts leaves boundary terms at  $t_s$ : arc 1 gives  $-\boldsymbol{\lambda}^\top(t_s^-) \delta \mathbf{z}(t_s^-)$  at its upper end, arc 2 gives  $+\boldsymbol{\lambda}^\top(t_s^+) \delta \mathbf{z}(t_s^+)$  at its lower end — opposite signs because  $t_s$  is the right edge of one arc and the left edge of the other. Since  $\mathbf{z}(t_s^-)$  and  $\mathbf{z}(t_s^+)$  are independent (the state jumps), the coefficient of each variation must vanish separately:

$$\begin{aligned} \text{coeff. of } \delta \mathbf{z}(t_s^-) : & \quad -\boldsymbol{\lambda}^\top(t_s^-) + \boldsymbol{\pi}^\top N_{\mathbf{z}^-} = 0, \\ \text{coeff. of } \delta \mathbf{z}(t_s^+) : & \quad +\boldsymbol{\lambda}^\top(t_s^+) + \boldsymbol{\pi}^\top N_{\mathbf{z}^+} = 0, \end{aligned} \quad (57)$$

and substituting  $N_{\mathbf{z}^\mp} = \mp I_5$  each collapses to  $\boldsymbol{\lambda}(t_s^-) = \boldsymbol{\lambda}(t_s^+) = -\boldsymbol{\pi}$ . Both sides equal the same constant, so the net jump is the sum of the two blocks,

$$\boldsymbol{\lambda}(t_s^-) - \boldsymbol{\lambda}(t_s^+) = \boldsymbol{\pi}^\top (N_{\mathbf{z}^-} + N_{\mathbf{z}^+}) = \boldsymbol{\pi}^\top (-I_5 + I_5) = \mathbf{0}. \quad (58)$$

**Every costate is continuous**, and the continuity is exactly the cancellation of the two blocks — which is why both are needed. The remaining coefficient, that of  $\delta t_s$ , gives the Hamiltonian jump of Step 5. In particular  $\lambda_m(t_s^-) = \lambda_m(t_s^+) \equiv \lambda_m(t_s)$ , the bilinear-tangent steering is unbroken, and only the mass steps, from  $m^- = 1 - Qt_s$  to  $m^+ = 1 - (1 + \eta)Qt_s$ .

**Why state-independence matters.** Had the stage shed a fraction of the vehicle instead,  $m^+ = (1 - k)m^-$ , then  $\partial m^+ / \partial m^- = 1 - k$ , the mass block of  $N_{\mathbf{z}}$  would no longer be  $\pm 1$ , and the  $\boldsymbol{\pi}^\top N_{\mathbf{z}}$  term would survive: the mass costate would jump,  $\lambda_m(t_s^-) = (1 - k) \lambda_m(t_s^+)$ . It is the absolute,  $m^-$ -independent jettison that lets  $\lambda_m$  pass through untouched.

**Step 4 — the Hamiltonian jump.** Because  $v_x, v_y$ , the velocity costates and  $\lambda_m$  are all continuous, the only discontinuous term in  $\mathcal{H}$  is the thrust acceleration  $(T/m)\|\boldsymbol{\lambda}_v\|$ , which sees the mass step:

$$\mathcal{H}(t_s^+) - \mathcal{H}(t_s^-) = T \|\boldsymbol{\lambda}_v(t_s)\| \left( \frac{1}{m^+} - \frac{1}{m^-} \right). \quad (59)$$

**Sign:**  $m^+ < m^- \Rightarrow 1/m^+ > 1/m^-$ , so the right-hand side is positive — the lighter post-staging vehicle has the larger thrust acceleration, and  $\mathcal{H}$  steps up.

**Step 5 — the junction-side jump (Eq. 3.5.5).** The same jump is fixed by the right-hand condition in (54). The only explicit  $t_s$  in the junction is the jettisoned mass, so  $N_t = \partial N / \partial t_s = (0, 0, 0, 0, \eta Q)^\top$ , and with  $\boldsymbol{\pi} = -\boldsymbol{\lambda}(t_s)$  from Step 3 its mass component is  $\pi_m = -\lambda_m(t_s)$ . Substituting into (3.5.5),  $\mathcal{H}(t_s^-) = \mathcal{H}(t_s^+) - \pi_m \eta Q$ , gives the jump directly:

$$\mathcal{H}(t_s^+) - \mathcal{H}(t_s^-) = -\eta Q \lambda_m(t_s). \quad (60)$$

It remains to pin down which  $\lambda_m$  this is. The staging objective is to maximize the payload (45), equivalently to minimize the total burn time  $t_f$ ; the final mass  $m(t_f)$  is left unconstrained (the payload is read off afterwards) and the cost  $\phi$  does not contain it. The terminal transversality condition  $\boldsymbol{\lambda}(t_f) = \partial \phi / \partial \mathbf{z}(t_f)$  then makes the mass costate vanish at burnout,

$$\lambda_m(t_f) = \left. \frac{\partial \phi}{\partial m} \right|_{t_f} = 0, \quad (61)$$

which is the natural scale of the minimum-time problem. Since  $\lambda_m(t_f) = 0$ , replacing  $-\lambda_m(t_s)$  by  $\lambda_m(t_f) - \lambda_m(t_s)$  adds a term that is exactly zero, so that (60) reads

$$\mathcal{H}(t_s^+) - \mathcal{H}(t_s^-) = \eta Q [\lambda_m(t_f) - \lambda_m(t_s)]. \quad (62)$$

The rewrite matters: it makes the condition independent of how the costates are scaled. The shooting BVP integrates  $\lambda_m$  from  $\lambda_{m,0} = 1$  at  $t = 0$ , not from  $\lambda_m(t_f) = 0$ ; because  $\dot{\lambda}_m = (T/m^2) \|\boldsymbol{\lambda}_v\|$  depends on the other costates and not on  $\lambda_m$  itself, the two scalings differ only by a constant offset of the whole  $\lambda_m(t)$  curve. That offset cancels in the difference  $\lambda_m(t_f) - \lambda_m(t_s)$  but survives in  $\lambda_m(t_s)$  alone, so (62) can be evaluated with whatever  $\lambda_m$  the BVP carries, whereas (60) could not. **Sign:**  $\lambda_m$  increasing  $\Rightarrow \lambda_m(t_f) > \lambda_m(t_s)$ , so the right-hand side is positive — consistent with the positive jump of Step 4. This is the Weierstrass–Erdmann corner condition in action [4, Sec. 3.13, Prob. 1]: with no explicit-time junction term both  $\boldsymbol{\lambda}$  and  $\mathcal{H}$  would be continuous, and it is precisely the explicit  $t_s$  of the jettisoned mass that lets  $\mathcal{H}$  jump while  $\boldsymbol{\lambda}$  stays continuous.

**Step 6 — equate and simplify.** Setting (59) equal to (62) and using  $T = cQ$ ,

$$\begin{aligned} T \|\boldsymbol{\lambda}_v\| \left( \frac{1}{m^+} - \frac{1}{m^-} \right) &= \eta Q [\lambda_m(t_f) - \lambda_m(t_s)] \\ \Rightarrow \quad \eta [\lambda_m(t_f) - \lambda_m(t_s)] &= c \|\boldsymbol{\lambda}_v(t_s)\| \left( \frac{1}{m^+} - \frac{1}{m^-} \right), \end{aligned} \quad (63)$$

which is the corner condition (47). Both sides are positive, so the equation is well posed.

**Step 7 — solve and verify.** Appending  $t_s$  as a fifth shooting unknown and this relation as a fifth residual, a single `fsolve` (`validate_staging_corner.m`) returns  $t_s^* = 0.336$ ,  $t_f = 0.424$ ,  $m_u = 0.068$  — identical to the parameter sweep of Table 5, with the residual norm at the  $10^{-10}$  level. The terminal-reference term  $\lambda_m(t_f)$  is essential: replacing  $\lambda_m(t_f) - \lambda_m(t_s)$  by  $\lambda_m(t_s)$  alone (the inner minimum-fuel costate, normalized at  $\lambda_{m,0} = 1$ ) misplaces the optimum to a spurious  $t_s \approx 0.225$ .

**Physical reading.** The condition is a marginal balance. The right-hand side is the propulsive benefit of staging: the jump  $1/m^+ - 1/m^-$  in thrust acceleration enabled by shedding the structure, priced by the velocity costate  $\|\boldsymbol{\lambda}_v\|$  (how valuable an increment of  $\Delta v$  is at that instant). The left-hand side is the structural penalty: a later staging carries  $\eta Q$  more inert mass per unit time, priced by the terminal-referenced mass costate. Staging too early throws away structure that still earns its keep; staging too late drags dead mass; the optimum equalizes the two, which is exactly  $dt_f/dt_s = 0$ .

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